1. Considere la sucesión definida recursivamente por:

$$\begin{cases} y_0 & = & 1 \\ y_n & = & \frac{1}{2} y_{n-1} + 3, \quad n \geq 1 \end{cases}$$

Entonces

a) Calcule y_3, y_4. (2 puntos)

b) Deduzca una fórmula explícita para y_n y compruebe que se puede escribir como:

$$y_n = \frac{1}{2^n} + 6\left(1 - \frac{1}{2^n}\right)$$

(Sug. use la fórmula para $\sum_{k=0}^{n} r^k$ que demostró en la Tarea 1). (6 puntos)

c) Calcule $\lim_{n \to \infty} y_n$ (2 puntos)

$n=0$

$y_0 = 1$

$n=1$

$$y_1 = \frac{1}{2} \cdot 1 + 3$$

$$y_2 = \frac{1}{2}\left(\frac{1}{2} \cdot 1 + 3\right) + 3$$

$$y_3 = \frac{1}{2} \cdot \left[\frac{1}{2}\left(\frac{1}{2} \cdot 1 + 3\right) + 3\right] + 3$$

$$y_4 = \frac{1}{2} \cdot \left[\frac{1}{2} \cdot \left[\frac{1}{2}\left(\frac{1}{2} \cdot 1 + 3\right) + 3\right] + 3\right] + 3$$

$$y_n = \frac{1}{2^n} + 6\left(1 - \frac{1}{2^n}\right) \quad \begin{cases} y_0 & = & 1 \\ y_n & = & \frac{1}{2} y_{n-1} + 3, \quad n \geq 1 \end{cases}$$

$n=1$ $$\frac{1}{2^1} + 6\left(1 - \frac{1}{2^1}\right) = \frac{1}{2} \cdot 1 + 3$$

$$\frac{7}{2} = \frac{7}{2} \quad \checkmark$$

$y_n = \frac{1}{2^n} + 6\left(1 - \frac{1}{2^n}\right)$ $\begin{cases} y_0 & = & 1 \\ y_n & = & \frac{1}{2}y_{n-1} + 3, \quad n \geq 1 \end{cases}$

$n = p \qquad y_p = \frac{1}{2^p} + 6\left(1 - \frac{1}{2^p}\right)$. H.I

$n = p+1 \qquad y_{p+1} = \frac{1}{2^{p+1}} + 6\left(1 - \frac{1}{2^{p+1}}\right) \longrightarrow \frac{1}{2^{p+1}} + 6 - \frac{6}{2^{p+1}}$

$y_{p+1} = \frac{1}{2} y_{(p+1)\cancel{-1}} + 3 \qquad \frac{1-6}{2^{p+1}} + 6$

$= \frac{1}{2} \cdot y_p + 3 \qquad$ Hi

$$\boxed{6 - \frac{5}{2^{p+1}}}$$

$\frac{1}{2^1}\left[\frac{1}{2^p} + 6\left(1 - \frac{1}{2^p}\right)\right] + 3$

$\frac{1}{2^{p+1}} + 3\left(1 - \frac{1}{2^p}\right) + 3$

$\frac{1}{2^{p+1}} + 3 - \frac{3}{2^p} + 3$

$\frac{1}{2^{p+1}} + 3 - \frac{3}{2^p} \cdot \frac{2}{2} + 3$

$\frac{1}{2^{p+1}} + 3 - \frac{6}{2^{p+1}} + 3$

$$\frac{1}{2^{p+1}} + 3 - \frac{6}{2^{p+1}} + 3$$

$$\frac{1}{2^{p+1}} + 6 - \frac{6}{2^{p+1}}$$

$$\frac{1}{2^{p+1}} - \frac{6}{2^{p+1}} + 6$$

$$\frac{1-6}{2^{p+1}} + 6$$

$$6 - \frac{5}{2^{p+1}}$$

$$\boxed{\frac{1}{2^{p+1}} + 6\left(1 - \frac{1}{2^{p+1}}\right)}$$ ✓

$$\lim_{n\to+\infty} \frac{1}{2^n} + 6\left(1 - \frac{1}{2^n}\right)$$

$$\overset{0}{\cancel{\frac{1}{+\infty}}} + 6\left(1 - \overset{0}{\cancel{\frac{1}{+\infty}}}\right)$$

$$\boxed{6}$$